

Self-falsifying quantum spectroscopy: a transportable necessary-condition screen for quantum-computed dynamical spectra

Nicolás Bonilla Vargas*

A quantum-computed dynamical spectrum sits at the sharp end of the verification paradox: the regime in which such a spectrum is scientifically most valuable—where classical simulation is intractable—is precisely the regime in which it can least be confirmed against a trusted reference [1, 2]. We propose treating a reconstructed spectrum not as an object to be certified but as a hypothesis to be exposed to refutation, and we equip it with a transportable battery of necessary-condition falsifiers. Each falsifier is a static ground-state spectral moment $m_k = \int \omega^k A(\omega) d\omega$ —for example $m_0 = \langle J^2 \rangle$ and $m_1 = \frac{1}{2} \langle [J, [H, J]] \rangle$ —computed by an estimator that is *independent* of the reconstruction, from the *same* computational-basis samples that produced the spectrum, and never read back off the reconstructed $A(\omega)$ [3–5]. Because the estimator is reconstruction-independent, a spectrum that agrees with its own moments is one that *could* have failed a decisive test and did not; this is Popperian corroboration, not proof [6]. We demonstrate the screen across distinct error classes and physical domains: (i) a spurious low-frequency (Drude-like) peak in the doped-Hubbard current spectral function that steals mid-infrared weight, preserving the total integrated weight $m_0 = \langle J^2 \rangle$ yet failing the first (mean-frequency) current moment m_1 [4, 7]; (ii) a make-or-break test in which the independent moment estimator fires on sample-based-diagonalization determinant-subspace truncation while a back-computed control is blind [2, 8]; and (iii) a cross-domain U(1) lattice-Schwinger check in which the exact Gauss-law constraint $[H, G] = 0$, with physical $\langle \sum G^2 \rangle = 0$, fires on gauge-breaking error [9, 10]. We demonstrate on real IBM Heron hardware using the sample-based-diagonalization spectral data of a companion study [5], and we prove a closed-form guarantee: under exact moments the extremal spread in cumulative spectral weight over all positive measures sharing $m_0, \dots, m_{2n}$ equals the Christoffel function $W_n(t) = 1 / \sum_{j=0}^n p_j(t)^2$ at interior points and bounds the deviation of any single reconstruction; it contracts monotonically with moment order [11–13]. We state the limits without hedging: the screen falsifies, it does not certify; its conditions are necessary, not sufficient; it targets one error class—subspace truncation / weight misassignment—not device noise in general [1]; it certifies cumulative/integrated weight only, not pointwise line shapes or peak positions; positivity of the underlying measure holds for diagonal $T = 0$ Lehmann spectra only [14]; and the exact-moment idealization must be replaced by a robust interval-moment version on hardware. Because the reconstruction is itself a learned generative model of the device’s samples, the screen doubles as a falsifiable-trust layer for machine-learned models of quantum-computed data—trust earned by surviving independent physical constraints rather than by fitting the samples—a prerequisite for using a quantum processor as an inferential and generative engine, and one asserted here with no accompanying claim of quantum learning advantage [6, 15]. The method is thereby complementary to, and mechanistically distinct from, self-verifying variational simulation [9] and observable estimation validated without a classical reference [1].

I. INTRODUCTION

Dynamical response functions are the quantities through which many-body theory meets experiment: the single-particle spectral function $A(k, \omega)$ is resolved by angle-resolved photoemission, the dynamical structure factors $S(q, \omega)$ and $S^{zz}(q, \omega)$ by inelastic neutron and resonant x-ray scattering, and the optical conductivity by infrared spectroscopy [16–20]. A quantum processor that computes these observables—rather than the static ground-state energies of the first generation of quantum simulation [21–25]—is producing the functions that carry the empirical content of correlated-electron physics, via sample-based diagonalization and quantum Krylov methods on superconducting and trapped-ion

hardware [2, 5, 8, 26–46]. Their scientific value is greatest exactly where a trusted classical cross-check is unavailable—the regime that motivates the quantum resource, and whose boundary is actively contested as once-“advantageous” demonstrations are dequantized by improved classical methods [47–52]. This is the verification paradox in its sharpest form: we most want to check a spectrum precisely where no independent standard is at hand to tell right from wrong [1, 53–57]. A computed spectrum earns the standing of physical knowledge, then, only once it can be held to the structure that every true spectral function of a Hamiltonian must possess irrespective of how it was obtained—the moment sum rules that fix its integrated weight and mean frequency [3, 4, 7], the conserved-charge identities its state cannot violate [10], and the positivity of the zero-temperature Lehmann measure [14]: not diagnostics of a particular algorithm, but the physics that survives the method.

The stakes are also architectural. A quantum-computed spectrum increasingly sits at the output end of

* M.Sc. Physics, Universidad Nacional de Colombia; M.Sc. Applied Data Science & AI, SRH University Heidelberg / Munich; Daita AI. ngbonillav@unal.edu.co

The self-falsification loop

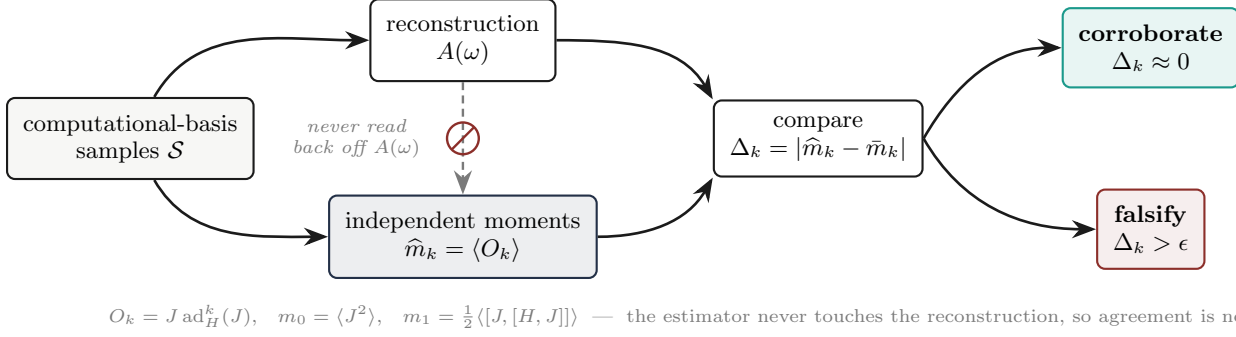


FIG. 1. **The self-falsification loop.** Computational-basis samples feed two branches that must agree: a spectral reconstruction $A(\omega)$ and a reconstruction-independent operator-moment estimator $\hat{m}_k = \langle O_k \rangle$, with $O_k = J \text{ad}_H^k(J)$ the probe times its k -fold commutator with H ($m_0 = \langle J^2 \rangle$, $m_1 = \frac{1}{2} \langle [J, [H, J]] \rangle$; Appendix A). A discrepancy $\Delta_k = |\hat{m}_k - \tilde{m}_k|$ beyond estimator uncertainty falsifies the spectrum, where $\tilde{m}_k = \int \omega^k A(\omega) d\omega$ is the moment read off the reconstruction. The estimator branch is drawn deliberately *not* reading from $A(\omega)$, making explicit that agreement was not preordained—the property that gives the screen teeth [3, 5, 6].

a quantum-centric supercomputing pipeline, in which the device supplies the samples and subspaces that classical resources then diagonalize and reconstruct [2, 5, 44, 46, 58–60], and the reconstruction step is naturally read as a learned generative decoder of the sampled distribution—a neural quantum state or a flow- or policy-based sampler [15, 61, 62], a landscape we survey elsewhere [63]. This is the sense in which the objective animating our programme comes into view: to use the quantum processor less as a solver of pre-specified numbers and more as an inferential and generative engine, a device whose sampled outputs are the data a learned model is built on. Such a use carries a prerequisite that is easy to defer, and which we take up here: it must be possible to *trust* the outputs. In the regime that motivates the machine, trust cannot be trust-by-fit—believing a reconstruction because it reproduces the samples it was trained on, which any internally consistent fit does equally well when no reference can catch a wrong one—but only trust-by-falsification: belief earned by surviving independent physical constraints the model was never fit to and could genuinely have failed [1, 6]. The order of dependence is what makes this foundational rather than ornamental: before a quantum processor can be argued to *learn or generate better* than its classical counterpart, what it generates must first be exposed to an independent test it could have failed—a falsifiable check of correctness, not a fit to its own samples—so that falsifiable correctness is logically prior to any claim of learning advantage, and it is this prior, more basic layer that the present work builds. Its emblematic instance is the single-particle spectral function of quantum matter (Fig. 3): a generative output that a wrong reconstruction can render indistinguishable from a valid one—to the eye and to a total-weight check alike—yet that a reconstruction-independent moment refutes, the clean

and minimal demonstration of the falsifiable-trust layer such a programme requires. What the pipeline still lacks is exactly such a check on the reconstructed spectral *function*: the number- and state-certifying guarantees of self-verifying state preparation [9], single-qubit-measurement state certification [64], and reference-free observable estimation [1] do not supply one. This paper supplies it—a transportable, reconstruction-independent falsifier attachable by any consumer of quantum-generated spectra—as the honest first component of the falsifiable-trust layer such a programme requires: it claims that role, and no quantum learning advantage beyond it, an advantage we find undemonstrated at the scales we can access [5].

Two influential responses to this predicament already exist, and it is essential to delineate them by mechanism rather than by aspiration. Self-verifying variational simulation prepares a target state variationally on the device [22, 65, 66] and certifies it by measuring an independent Hamiltonian-derived quantity—an energy variance—on that same state, bounding its distance to a true eigenstate [9]. A distinct line of work establishes trust in error-mitigated observable estimates by validating the hardware noise model and the mitigation procedure [67–71], or by predicting many observables from randomized measurements without full tomography [72–77], without recourse to a classical reference [1]. Both approaches certify a *number*—an energy or an expectation value—and both are powerful within that remit [1, 9]. Neither is designed to interrogate the internal self-consistency of a reconstructed *function* such as a full $A(\omega)$, and neither screens the characteristic failure mode of subspace-based spectral methods, in which spectral weight is placed at the wrong frequencies while integrated quantities can appear healthy—a mode now documented in critical studies of quantum-selected con-

figuration interaction and quantum subspace diagonalization [5, 40, 78–80].

We take a different epistemic stance, borrowed directly from the logic of scientific discovery: a claim earns credence not by being provable but by surviving genuine opportunities to be refuted [6]. Falsifiability, not verifiability, is the workable demarcation in a regime where confirmation is out of reach [6]. We therefore convert an unconfirmable spectrum into a corroborated one by attaching to it a battery of necessary-condition falsifiers: tests that the true spectrum *must* pass, so that a failure is logically decisive while a success is corroboration and never proof [6]. The value delivered is precise and modest: a spectrum equipped with such a screen is one that could have failed a real test and did not [6].

The falsifiers are spectral moments $m_k = \int \omega^k A(\omega) d\omega$. For a zero-temperature Lehmann spectral function these moments coincide with static ground-state expectation values— $m_0 = \langle J^2 \rangle$, $m_1 = \frac{1}{2} \langle [J, [H, J]] \rangle$, and so on—obtainable directly from operator algebra rather than from the dynamics [14, 81]. The double-commutator form of these moment/sum-rule identities is standard [3]. In the optical-conductivity setting $m_0 = \langle J^2 \rangle$ is the total integrated current-current weight, whereas the optical f -sum rule $\int_0^\infty \sigma_1 d\omega = \frac{\pi}{2} \langle -\hat{T} \rangle$ is the distinct m_{-1} moment—a ground-state kinetic energy whose character was established by Kohn and made explicit for the Hubbard chain by Maldague [4, 7, 16]. The question of which moment sequences $\{m_k\}$ are consistent with a positive spectral measure, and of how tightly finitely many moments constrain that measure, is the classical moment problem, with a mature theory of existence, determinacy, and extremal bounds [82–85]. The bridge from moments to orthogonal polynomials and Gauss quadrature—the machinery underlying our closed-form guarantee—is likewise classical [11, 12, 86–88], and is the same machinery that powers the kernel-polynomial and recursion methods routinely used to compute spectra on classical hardware [89–92].

The decisive design choice is *independence*. Each moment in our screen is computed by an estimator that does not touch the reconstruction: it is evaluated as an operator expectation value on the same computational-basis samples, not by integrating the reconstructed $A(\omega)$ [5]. This distinction is the whole point. A moment read back off the reconstruction is circular—it is guaranteed to agree and therefore can never falsify—whereas an independent estimator built from operator algebra can genuinely disagree with the reconstruction and thus can fire. The screen is, in this sense, a self-falsification device (Fig. 1): the spectrum is confronted with quantities that it did not itself generate [6]. This reading has a direct machine-learning counterpart, developed in the Discussion: because the reconstruction is itself a learned generative model of the samples, the screen is equivalently a necessary-condition falsifier for that model—trust earned by falsification, not by fit [6].

We instantiate the screen across error classes and phys-

ical domains to show that it is transportable rather than tuned. In the doped Hubbard current response, a reconstruction that invents a spurious low-frequency (Drude-like) peak, misplacing mid-infrared weight to low energy, preserves the total integrated weight $m_0 = \langle J^2 \rangle$ yet fails the first (mean-frequency) current moment m_1 , exposing a wrong distribution of a correct total weight [4, 7]. In sample-based diagonalization, determinant-subspace truncation is detected by the independent moment estimator while a back-computed control remains blind—a make-or-break demonstration of why circularity must be avoided [2, 8]. Crossing domains, a U(1) lattice-Schwinger check exploits the exact Gauss-law constraint $[H, G] = 0$, whose physical value $\langle \sum G^2 \rangle$ vanishes, to fire on a genuinely different error class—gauge-symmetry breaking—an error type whose measurability was established experimentally in an analog cold-atom platform [10] and whose digital lattice-Schwinger context is the model treated by self-verifying simulation [9]. We then demonstrate the screen on real IBM Heron hardware using the sample-based-diagonalization spectral data of the companion study [5], and we close the loop with a theorem: under exact moments the extremal spread in cumulative spectral weight equals the Christoffel function $W_n(t) = 1/\sum_{j=0}^n p_j(t)^2$ at interior points, a certified quantity that contracts monotonically as more moments are enforced [11, 13].

We are deliberate about scope from the outset, because an honest screen is more useful than an overclaimed one. The method falsifies but does not certify; its conditions are necessary but not sufficient; it screens one error class—subspace truncation and weight misassignment—and not device noise in general, which remains the province of noise-model and observable-estimation validation [1]. It certifies integrated weight, not pointwise line shapes; its positivity structure is that of a diagonal $T = 0$ Lehmann spectrum [14]; and its exact-moment idealization must be softened to robust interval moments before it is trustworthy on noisy hardware. These limits are stated here, carried through the paper, and revisited in the Discussion, not tucked away.

II. METHOD: RECONSTRUCTION-INDEPENDENT MOMENT FALSIFIERS

We formalize the screen in two steps: first the object it tests—a quantum-computed spectrum and the samples that produced it—and then the falsifiers we attach to it.

A. The quantum processor and the origin of the samples

The object the screen tests is produced by a definite quantum-computational pipeline, and the screen acquires meaning only once that pipeline is fixed. The proces-

sor’s task, inherited from the companion study [5] and its sample-based-diagonalization lineage [2, 8, 44], is not to evaluate $A(\omega)$ directly but to *define, by physical sampling, the correlated subspace* in which the spectrum is then built by exact classical diagonalization. A correlated input state is prepared on an IBM Heron processor—a hardware-native local unitary cluster-Jastrow ansatz in the chemistry setting [2], or a Trotterized real-time evolution of an $(N\pm 1)$ -particle seed $c_p^\dagger|\psi_0\rangle$ in the dynamical setting of the companion study [5, 30, 42]—and measured shot by shot in the occupation-number basis. Each shot returns a bitstring, one candidate Slater determinant, whose empirical frequency estimates $|\langle\text{config}|\psi\rangle|^2$ [8]. The distinct measured bitstrings, after symmetry-restoring configuration recovery to the correct particle-number and spin sector, form the subspace $\mathcal{S}$; the Hamiltonian is projected onto $\mathcal{S}$ and diagonalized classically, and the dynamical spectral function is assembled as the $T = 0$ Lehmann sum $A(\omega) = \sum_n |\langle n|J|0\rangle|^2 \delta_\eta(\omega - (E_n - E_0))$ over the resulting eigenpairs, with δ_η a narrow broadening [5, 14].

The processor’s role is essential in principle at scale—though not at the classically reproducible sizes we demonstrate, where the full sector is recovered by coverage: the exact many-body sector grows exponentially with system size, so the configurations that carry spectral weight cannot in general be enumerated classically, and the correlated prepared state acts as a physically informed importance sampler that concentrates shots on exactly those configurations—so that $\mathcal{S}$ is fixed by measurement rather than by a classical selection heuristic [2, 5]. This is precisely what makes the screen non-trivial rather than circular: the reconstruction-independent moment estimators $\hat{m}_k = \langle O_k \rangle$ defined next are evaluated on the *same* measured bitstrings $\mathcal{S}$, and inherit the same shot noise and error-mitigation choices as the spectrum they test.

B. Reconstruction-independent moment falsifiers

A quantum-computed dynamical spectrum is thus delivered as a reconstructed non-negative function $A(\omega)$ together with the sample set $\mathcal{S}$ just described [5]. For a zero-temperature diagonal Lehmann spectral function of an operator J , $A(\omega) = \sum_n |\langle n|J|0\rangle|^2 \delta(\omega - (E_n - E_0))$ is a positive measure on the real line, and its power moments $m_k = \int \omega^k A(\omega) d\omega$ are ground-state expectation values of operators built from nested commutators of J with H : $m_0 = \langle 0|J^2|0\rangle$, $m_1 = \frac{1}{2}\langle 0|[J, [H, J]]|0\rangle$, and in general $m_k = \langle O_k \rangle$ with $O_k = J \text{ad}_H^k(J)$ the probe times its k -fold commutator with H , derived in Appendix A [3, 14]. This is the standard moment/sum-rule construction [3]; in the optical case $m_0 = \langle J^2 \rangle$ is the total integrated weight and m_1 its mean frequency, while the optical f -sum is the separate m_{-1} moment [4, 7].

The screen attaches to the spectrum an *independent* estimator $\hat{m}_k$ of each moment. Independence has an

operational definition: $\hat{m}_k$ is computed as the sample estimate of $\langle O_k \rangle$ over $\mathcal{S}$ —via the same diagonal-in-the-computational-basis or importance-weighted estimators used to obtain ground-state observables in sample-based diagonalization—and is at no point obtained by numerically integrating $\omega^k A(\omega)$ [2, 5, 8]. We denote by $\bar{m}_k = \int \omega^k A(\omega) d\omega$ the reconstruction moment, the quantity a circular check would use. The falsifier is the discrepancy $\Delta_k = |\hat{m}_k - \bar{m}_k|$: the true spectrum must satisfy $\Delta_k = 0$ up to statistical error, so a discrepancy exceeding the estimator’s uncertainty is a decisive refutation. Crucially, a circular check compares $\bar{m}_k$ to itself and yields $\Delta_k \equiv 0$ by construction, and therefore can never fire; the make-or-break Results section demonstrates this contrast on real truncation error.

The screen is a battery, not a single test: it enforces Δ_k for $k = 0, 1, \dots, K$ and, separately, the moment-problem consistency of the sequence $\{\hat{m}_k\}$ itself—the Hankel positive-semidefiniteness that any sequence of moments of a positive measure must satisfy [82, 83]. A violated Hankel condition falsifies the spectrum even without reference to $A(\omega)$ —a moment sequence is thus refutable by geometry alone (Fig. 2)—because no positive measure, hence no physical $T = 0$ Lehmann spectrum, has those moments [83, 84]. Because a diagonal $T = 0$ Lehmann measure is supported on $\omega = E_n - E_0 \geq 0$, the sharp condition is in fact the Stieltjes pair $H_n = [m_{i+j}] \geq 0$ and the shifted $H_n^{(1)} = [m_{i+j+1}] \geq 0$; the shifted matrix is an additional native falsifier that a bare Hankel test discards. The battery also accepts any exactly known operator identity as a native falsifier: a conserved charge $[H, G] = 0$ with a known physical eigenvalue supplies a moment-like constraint whose violation is likewise decisive, as we exploit in the gauge-theory domain [10]. The construction is transportable because it depends only on (a) a positive spectral measure and (b) operators whose ground-state expectations are computable from the same samples—both of which hold across the domains we treat [11, 89].

III. RESULTS I: TWO DOMAINS, TWO ERROR CLASSES

We begin by displaying the physical spectra on which the screen operates. Figure 3 shows the single-particle spectral function $A(k, \omega)$ of the half-filled Hubbard ring, with its lower and upper Hubbard bands separated by the Mott gap and the characteristic dispersion of a one-dimensional correlated insulator [5, 17, 93–96]. A truncated or otherwise corrupted reconstruction misplaces spectral weight—here, the reconstructed Hubbard-band centroid dispersion is flattened by a momentum-resolved inter-band weight misassignment that leaves the Mott gap and the per- k total weight intact, the single-particle, momentum-resolved counterpart of the optical weight-misplacement of Fig. 5 in the weight-misassignment family quantified in Fig. 6—and the reconstruction-

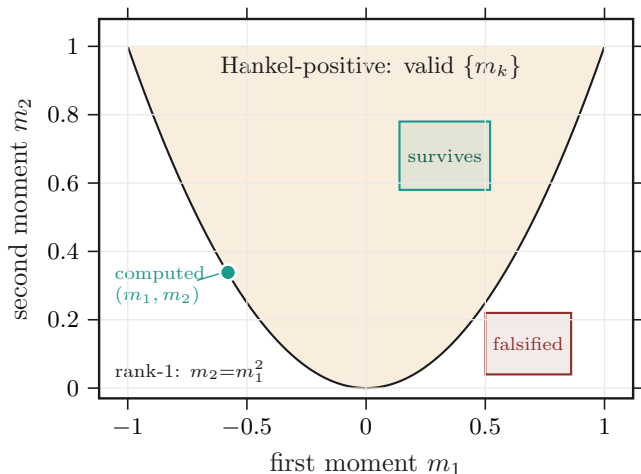


FIG. 2. **A moment sequence is falsifiable by geometry alone.** In the (m_1, m_2) plane (normalized measure rescaled to $[-1, 1]$), a genuine spectrum's moments must lie in the Hankel-positive body $m_1^2 \leq m_2 \leq 1$; the lower boundary $m_2 = m_1^2$ is the rank-one limit of a single δ . A candidate whose shot-budget interval box lies entirely below the rank-one boundary—wholly in the Hankel-negative region $m_2 < m_1^2$ (oxblood)—is refuted with no reference to $A(\omega)$, whereas one that lies inside the body (teal) survives. The real computed moments of the doped-Hubbard current spectrum (teal dot) sit essentially on the boundary—the measure is dominated by one spectral atom, hence nearly rank-one [82, 83].

independent first moment, evaluated at each momentum k via the anticommutator sum rule appropriate to the non-Hermitian single-particle probe $c_{k\sigma}$ (Appendix A), detects the misplacement (right panel). The same construction applies to the collective response: Figure 4 shows the charge dynamical structure factor $S(q, \omega)$, gapped by the Mott scale Δ_c , alongside the spin structure factor $S^{zz}(q, \omega)$, whose gapless low-energy weight is the spinon continuum of the underlying Luttinger liquid—the hallmark of spin-charge separation [3, 5, 17–19]. Each channel is a positive $T = 0$ Lehmann measure carrying its own moment sum rules, so the screen transports across observables as well as across models. Such spectra are the benchmark output of mature classical many-body methods—density-matrix renormalization group and tensor-network dynamics, dynamical mean-field theory, and determinant quantum Monte Carlo with analytic continuation—against whose established features the moment screen's necessary conditions can themselves be checked [97–106].

We now exercise the screen on a wrong distribution of correct total weight. In the doped one-dimensional Hubbard model the optical conductivity carries a coherent Drude weight at $\omega = 0$ (the charge stiffness) together with a mid-infrared/Mott band that dominates the regular ($\omega > 0$) spectrum, and the total integrated current-current weight is fixed by the zeroth moment $m_0 = \langle J^2 \rangle$ —a ground-state quantity [4, 7, 16, 108]. We

construct a wrong reconstruction that invents a spurious low-frequency (Drude-like) peak, misplacing a fraction of the mid-infrared weight to low energy while preserving the total. Figure 5 shows the outcome: the zeroth moment m_0 (the total integrated weight) is satisfied to within estimator error, so a total-weight check alone passes the wrong spectrum; the first current moment $m_1 = \frac{1}{2} \langle [J, [H, J]] \rangle$, computed independently from the samples, disagrees with the reconstruction's first moment by many times its uncertainty, and the spectrum is refuted [3, 4]. The lesson is that integrated total weight is a weak falsifier and that the first moment—the mean frequency of the weight—is precisely the quantity a misplaced-weight reconstruction corrupts.

We then cross domains to a different error class entirely. In the $U(1)$ lattice Schwinger model the generator of gauge transformations G_x commutes with the Hamiltonian, $[H, G] = 0$, and every physical state lies in the $G_x = 0$ sector, so $\langle \sum_x G_x^2 \rangle = 0$ exactly [10, 109–111]. This is a native, exact falsifier that requires no moment reconstruction: a computed state or spectrum contaminated by gauge-symmetry-breaking error acquires $\langle \sum_x G_x^2 \rangle > 0$, and the screen fires (Fig. 7). The measurability of Gauss-law violation as a diagnostic was established experimentally in an analog cold-atom quantum simulator [10, 112], and the digital lattice-Schwinger model itself is the system on which self-verifying variational simulation [9] and early digital gauge-theory dynamics [113, 114] were demonstrated; our contribution is to cast the Gauss law as a same-sample necessary-condition screen on a computed spectrum rather than as a state-preparation diagnostic. Together the two domains show that the screen is not tuned to one physical setting: a weight-misassignment error in a correlated-electron spectrum and a symmetry-breaking error in a gauge theory are caught by the same construction, each with the falsifier appropriate to its physics [10, 89].

IV. RESULTS II: THE TEETH—INDEPENDENT ESTIMATOR VS CIRCULAR CONTROL

The make-or-break claim of the method is that *independence* is not a stylistic preference but the property that gives the screen teeth. We test it on the native failure mode of sample-based diagonalization: determinant-subspace truncation, in which the sampled subspace omits configurations that carry real spectral weight, so the reconstructed spectrum is internally consistent with an incomplete basis [2, 8]. We run two checks side by side on the same truncated data (Fig. 6). The independent estimator computes $\hat{m}_k = \langle O_k \rangle$ from an operator expectation that references the full operator algebra, not only the retained determinants; the back-computed control computes $\bar{m}_k$ by integrating the reconstructed $A(\omega)$, which lives entirely inside the truncated subspace. As truncation is tightened, the independent estimator's discrepancy Δ_k grows and crosses the falsification threshold,

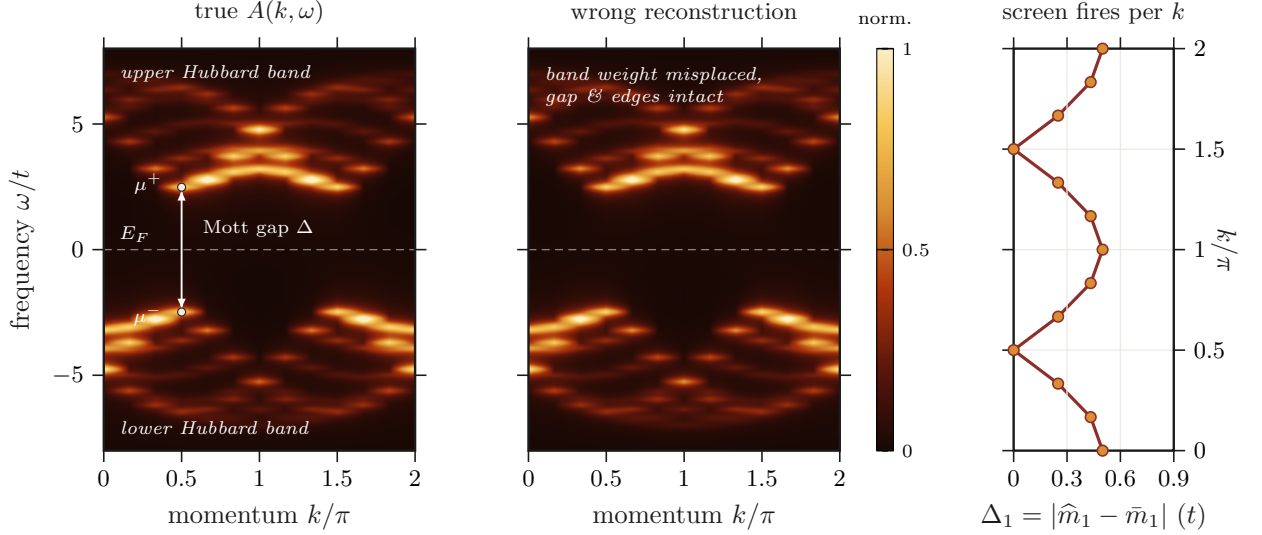


FIG. 3. **The self-falsifying screen on the single-particle spectral function $A(k, \omega)$ of the Hubbard model.** Left: the true $A(k, \omega)$ of the half-filled Hubbard ring, with the Fermi level E_F at mid-gap and the dispersing upper/lower Hubbard bands (UHB/LHB) separated by the Mott gap; the correlation-driven edges $\mu^\pm \approx \pm 2.484t$, marked by the dots on the band edges at $k_F = \pi/2$ and spanned by the arrow, give the single-particle Mott gap $\Delta = \mu^+ - \mu^- \approx 4.97t$ (approaching U only in the atomic limit $t \rightarrow 0$). Centre: an illustrative injected corruption, modeled on the determinant-truncation weight-misassignment mode quantified in Fig. 6: at each momentum a fraction of upper-Hubbard-band (electron-addition) intensity is rebalanced against the lower band, flattening the reconstructed band-centroid dispersion while leaving the Mott gap strictly empty ($A(k, E_F) = 0$), the edges $\mu^\pm$ fixed, and the per- k total weight $m_0 = \langle \{c_{k\sigma}, c_{k\sigma}^\dagger\} \rangle = 1$ exactly preserved (to machine precision)—a Mott spectrum that a casual look and a total-weight check both accept, indistinguishable from the true panel by eye. Right: the reconstruction-independent first moment $\hat{m}_1(k) = \langle \{[c_{k\sigma}, H], c_{k\sigma}^\dagger\} \rangle = \epsilon_k + U\langle n_{-\sigma} \rangle$, the Hartree-shifted band centroid computed per momentum from the ground state (Appendix A), departs from the reconstruction’s $\bar{m}_1$ by $\Delta_1 = |\hat{m}_1 - \bar{m}_1| \propto |\epsilon_k|$: the screen fires hardest where the bare dispersion is largest in magnitude, at the band extrema $k = 0, \pi, 2\pi$ ($\Delta_1 = 0.50t$), and vanishes exactly at the native momenta $k = \pi/2, 3\pi/2$ where $\epsilon_k = 0$ —a self-consistency point no flat-dispersion error can hide behind. For this single-particle probe m_0 is pinned to 1 by the equal-time anticommutator and carries no discriminating power (blind), so the first moment is the first non-trivial falsifier [3, 5, 14, 107]. Sector-basis Lanczos ground state with Haydock continued-fraction Green’s functions [88, 90], $L = 12$ sites at half filling (12 exact momenta), $U/t = 8$, broadening $\eta = 0.16t$; the Hubbard bands now disperse cleanly, while spinon/holon fractionalization remains the province of the collective response of Fig. 4.

while the circular control’s discrepancy stays identically zero and reports the spectrum as healthy [2]. The circular control is blind by construction: it can only ever confirm that the reconstruction is consistent with itself. This is the concrete demonstration behind the Method’s warning, and it is the reason every moment in the screen must be estimated off the samples and never read back off $A(\omega)$ [5, 8].

V. A CLOSED-FORM GUARANTEE: THE CHRISTOFFEL MISS-DISTANCE

The screen also carries a quantitative guarantee, drawn entirely from the classical theory of moments and orthogonal polynomials; the novelty is its deployment as a certificate for a quantum-computed spectrum, not the mathematics—the Christoffel–Darboux kernel is by now an established data-analysis certificate in its own right [11, 12, 115, 116]. Consider the cumulative spectral weight $F(t) = \int_{-\infty}^t A(\omega) d\omega$ of a diagonal $T = 0$

Lehmann measure, and suppose the first $2n+1$ moments $m_0, \dots, m_{2n}$ are known exactly, which fix the orthonormal polynomials $p_0, \dots, p_n$ via the Hankel system [82]. Let $W_n(t) = 1/\sum_{j=0}^n p_j(t)^2$ be the Christoffel function, also written $\lambda_n(t) \equiv W_n(t) = 1/K_n(t, t)$ with $K_n(x, y) = \sum_{j=0}^n p_j(x)p_j(y)$ the Christoffel–Darboux kernel; its explicit construction from the moments is given in Appendix D [11, 12].

Theorem 1 (Certified cumulative-weight miss-distance). *Let $d\mu = A(\omega) d\omega$ be a positive measure of compact support (a diagonal $T = 0$ Lehmann spectrum) whose Hankel matrix $H_n = [m_{i+j}]_{i,j=0}^n$ is positive definite, and suppose the moments $m_0, \dots, m_{2n}$ are known exactly. Define the cumulative-weight miss-distance at t as the extremal spread $\sup\{|F_1(t) - F_2(t)|\}$ over all positive measures μ_1, μ_2 sharing those moments. Then for t in the interior of $\text{supp } \mu$ this extremal spread equals the Christoffel function $W_n(t) = 1/\sum_{j=0}^n p_j(t)^2$ —the maximal mass a moment-matching measure can place at t , i.e. the Gauss–Radau (Christoffel) weight $\lambda_n(t)$; for any sin-*

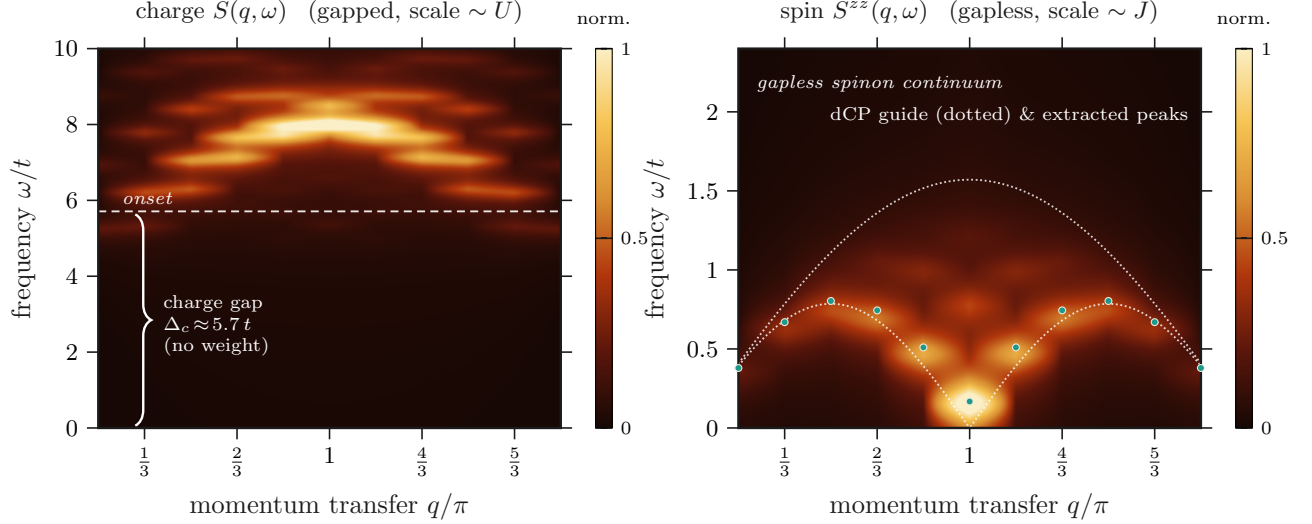


FIG. 4. **Spin–charge separation, and the breadth of the screen across observables.** The charge dynamical structure factor $S(q, \omega)$ (left) is gapped by the charge (density) onset $\Delta_c \approx 5.7t$ —the particle–hole excitation gap, a distinct finite-size quantity from the single-particle Mott gap $\mu^+ - \mu^- \approx 4.97t$ of Fig. 3, both set by the Hubbard scale U and coinciding only in the thermodynamic limit—while the spin structure factor $S^{zz}(q, \omega)$ (right) carries gapless low-energy spinon weight, the hallmark of the one-dimensional Luttinger liquid. The teal curves are the des Cloizeaux–Pearson (dCP) two-spinon boundaries $\omega_L = (\pi J/2)|\sin q|$, $\omega_U = \pi J|\sin(q/2)|$ with $J = 4t^2/U = 0.5t$, and the open points are the dominant spinon peak extracted per momentum, which tracks the lower dCP edge—an ideal large- U Heisenberg guide over the eleven momenta $q/\pi \in \{1/6, \dots, 11/6\}$, not a strict bound (finite U , finite size and broadening $\eta = 0.12t$ push some weight below the ideal lower boundary; the field is linearly interpolated in q for display). Both are positive $T = 0$ Lehmann measures with their own moment sum rules, so the same reconstruction-independent screen applies to each [3, 5]. The two panels are drawn on different frequency windows— $\omega \lesssim U$ for the charge, $\omega \lesssim$ a few $J = 4t^2/U$ for the spin—because the two channels’ excitations live on parametrically separated energy scales; that separation is itself the visual signature of spin–charge separation, and it is why a single shared axis would render the spinon continuum illegible. Sector-basis Lanczos ground state with Haydock continued-fraction evaluation [88, 90], $L = 12$ sites at half filling (11 momentum transfers), $U/t = 8$; broadening $\eta = 0.20t$ (charge) and $0.12t$ (spin).

gle reconstruction consistent with the moments the deviation from the true $F(t)$ is bounded above by $W_n(t)$ everywhere (proved in Appendix C). The certificate is monotone, $W_{n+1}(t) \leq W_n(t)$, since $K_{n+1} \geq K_n$. For a fixed finite Lehmann spectrum of N poles the construction is defined only for $n \leq N - 1$; at a pole ω_i one then has $W_n(\omega_i) \downarrow w_i > 0$ (the residue), so W_n does not vanish on the support. Pointwise vanishing $W_n(t) \rightarrow 0$ is instead a thermodynamic-limit statement: if the discrete spectrum densifies to an absolutely continuous density A on a Stahl–Totik regular support S obeying the local Szegő condition $\int_I \log A > -\infty$, then $nW_n(x) \rightarrow A(x)/\omega_S(x)$ for almost every $x \in I$, with ω_S the equilibrium density of S [13], reducing on a single interval to Máté–Nevai–Totik [117].

The extremal ordinate spread at t equals the maximal atom a moment-matching measure can carry at t —the Gauss–Radau (Christoffel) weight $\lambda_n(t) = W_n(t)$ —which is the classical content of the Chebyshev–Markov–Stieltjes inequalities and Gauss–Christoffel moment theory, proved in Appendix C [11, 82, 83]. What this buys the screen is a certified, computable resolution. Passing the moment battery through order $2n$ certifies the cu-

mulative weight $F(t)$ to within $W_n(t)$, a number we can evaluate, and enforcing more moments provably shrinks it [11, 13]. The boundary of the claim is equally explicit: the certified object is cumulative, integrated weight, not the pointwise value of $A(\omega)$ nor the position of any peak [11]; and the contraction rate is conditional on the regularity hypothesis and is not unconditional [13, 117]. Figure 8 makes the mechanism visible: the true cumulative weight $F(t)$ lies trapped inside the Gauss–Radau extremal envelopes over all measures sharing the moments, and the width of that bracket is exactly $W_n(t)$, which narrows as n grows. Figure 9 shows $W_n(t)$ itself contracting with n on the same measure.

VI. DEMONSTRATION ON IBM HERON HARDWARE

Finally we run the screen on real hardware, on the sample-based-diagonalization spectral data measured on an IBM Heron processor in a companion study, following exactly the pipeline of Sec. II A [5]. The reconstruction is the dynamical spectral function produced there by bitstring-sampled quantum subspaces; the mo-

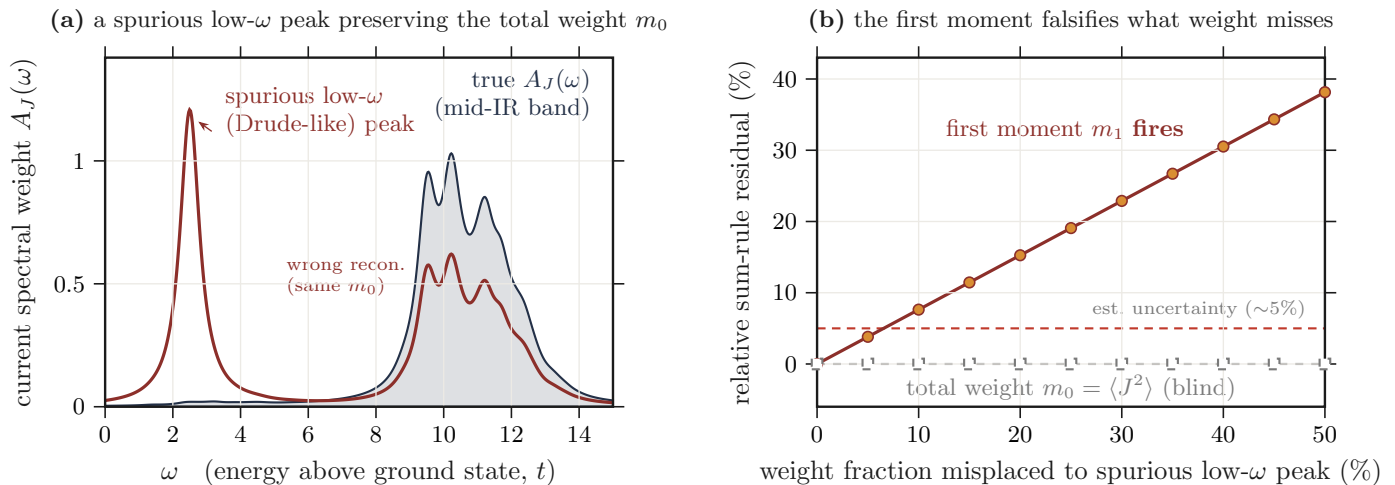


FIG. 5. **A wrong distribution of correct total weight.** A wrong reconstruction of the doped one-dimensional Hubbard current spectral function $A_J(\omega)$ (whose optical conductivity is $\text{Re}\sigma(\omega) \propto A_J(\omega)/\omega$) that invents a spurious low-frequency (Drude-like) peak, stealing weight from the mid-infrared/Mott band that carries essentially all the regular optical weight, *passes* the total integrated weight (zeroth moment $m_0 = \langle J^2 \rangle$) yet *fails* the first current moment $m_1 = \frac{1}{2} \langle [J, [H, J]] \rangle$ (wrong mean frequency of the weight), computed independently from the samples. **(a)** the true mid-IR band vs the wrong reconstruction (equal m_0); **(b)** as more weight is misplaced, the zeroth moment stays blind while the first moment fires. Sector-basis Lanczos ground state with Haydock continued-fraction / Ritz evaluation [88, 90], doped $L = 12$ ring at filling $2/3$, $U/t = 8$ [3, 4, 7].

ment estimators are computed from the same measured bitstrings, and are therefore subject to the same shot noise and error-mitigation choices as the spectrum they test [2, 5]. Figure 10 shows the screen firing correctly on the hardware data: the reconstruction-independent moments agree with the reconstruction within the retained-subspace regime and disagree beyond it, reproducing on real data the teeth-vs-blind contrast established in Results II [5, 8]. We stress the honest reading of this demonstration. The hardware moments are not exact; they carry statistical and systematic error, and the exact-moment theorem of the previous section is an idealization that a robust interval-moment version must recover before the Christoffel guarantee can be quoted on hardware [1]. What the Heron demonstration establishes is only that the construction is executable on a real device and fires on a deliberately induced error—not that any pointwise line shape has been certified [5]. As a bridge to a device-side deployment, a *classical* shot-budget forecast of the robust interval-moment test—not a device measurement, but a model on the exact state that fixes the falsification threshold from the real `ibm_fez` budget rather than by hand—is given in Appendix B.

VII. DISCUSSION AND LIMITATIONS

The contribution of this work is a change of stance more than a new device: a quantum-computed spectrum should be shipped not as a certified answer but as a falsifiable hypothesis, accompanied by the tests it was given the chance to fail [6]. What our screen delivers is cor-

roboration in the strict sense—a spectrum that survived reconstruction-independent necessary-condition checks—and never proof of correctness [6]. This is the honest reading of a necessary condition: passing raises credence and pinpoints the absence of a specific failure mode, while failing is decisive [6, 82]. The epistemic gain is real precisely because the estimator is independent of the reconstruction, so that agreement was not preordained [5].

The method must be positioned by mechanism against the two neighbouring lanes, because superficially all three verify quantum results without a classical reference. Self-verifying variational simulation certifies a prepared *state* via an energy-variance bound measured on the device [9]; observable-estimation validation certifies error-mitigated *expectation values* by validating the hardware noise model and mitigation procedure without a classical reference [1]. Our screen does neither: it certifies neither a prepared state nor an expectation value, and instead tests the internal consistency of a reconstructed spectral *function* against operator moments computed from the same samples, targeting the weight-misassignment failure mode of subspace methods that the other two lanes do not address [2, 5]. The lanes are complementary—a hardware spectrum could, in principle, be produced under observable-estimation validation [1], prepared with self-verification of its input state [9], and then screened for weight misassignment by our moments—and we claim only the third role.

The closed-form guarantee sets the precise currency of the screen. The certified miss-distance in cumulative spectral weight is the Christoffel function $W_n(t) = 1/\sum_{j \leq n} p_j(t)^2$, a classical object of the

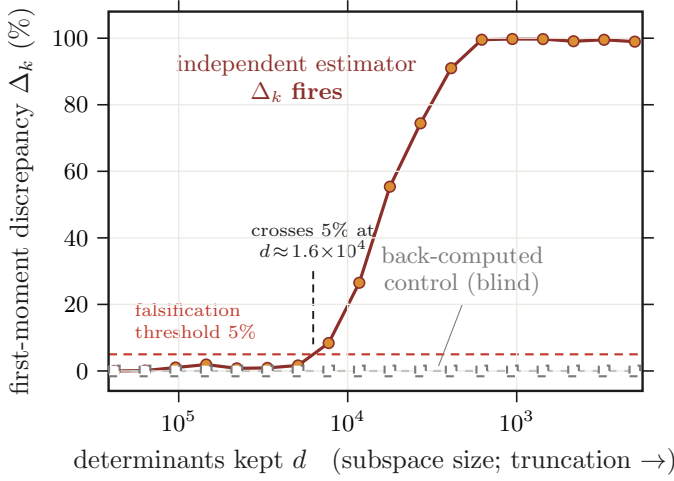


FIG. 6. **Make-or-break: independence gives the screen teeth.** The determinant subspace of the doped $L = 12$ Hubbard ring is truncated to its top d configurations by $|\psi_0|^2$ (the sample-based-diagonalization ordering; full support $n_{\text{sup}} = 2.4 \times 10^5$), and the current first moment is recomputed within it. On this truncated data the reconstruction-independent ground-state estimator’s discrepancy $\Delta_1^{\text{rel}} = |\hat{m}_1 - \bar{m}_1|/\hat{m}_1$ stays negligible while the subspace is large but grows sharply once truncation bites, crossing the falsification threshold at $d \approx 1.6 \times 10^4$ and saturating near 100%; a back-computed (circular) control read off the reconstructed spectrum stays identically zero and reports it healthy. The independent estimator therefore has teeth on the subspace-truncation error class—and, honestly, the first moment is *robust* until the subspace is aggressively small, so the screen catches severe truncation, not mild undersampling. The circular control is blind by construction [2, 5, 8]. Sector-basis Lanczos: the truncated first moment is $\langle J\psi_0^d | (H_d - E_0^d) | J\psi_0^d \rangle$ from one subspace ground state, no full diagonalization [88].

moment/orthogonal-polynomial theory, and it shrinks as the number of enforced moments grows [11, 12]. Its asymptotic contraction rate is governed by the $n \lambda_n$ behaviour established by Totik for Christoffel functions of regular measures on the real line [13]—reducing on a single interval to Máté–Nevai–Totik [117]—which ties the resolution of our bound to well-understood approximation theory rather than to a heuristic. This is also the boundary of the claim: the certified quantity is cumulative, integrated weight, not the pointwise value of $A(\omega)$ nor the position of any individual peak [11]. A screen that passes tells us the coarse-grained weight is placed consistently; it does not resolve a line shape [11, 89].

We restate the limits plainly, since overstating them would defeat the paper’s purpose. First, necessary is not sufficient: infinitely many wrong spectra share any finite set of moments, so passing the screen can never certify correctness [82, 83]. Second, the screen addresses one error class—subspace truncation and weight misassignment—and is not a general noise diagnostic; device-noise trust remains the domain of observable-

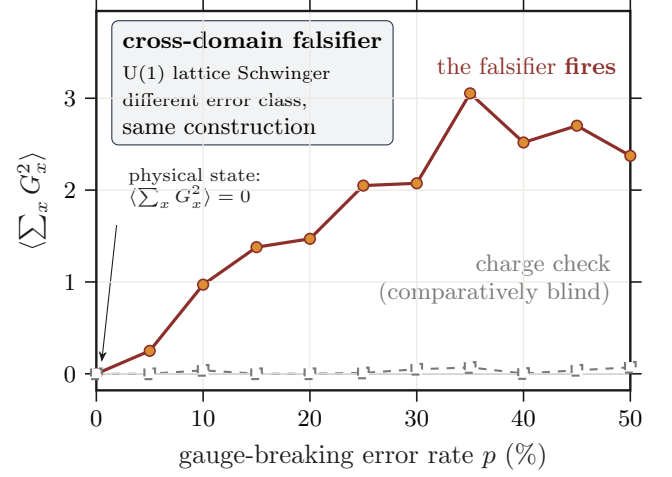


FIG. 7. **Cross-domain, different error class.** In the $U(1)$ lattice Schwinger model the exact Gauss-law constraint $[H, G] = 0$ places every physical state in the $G_x = 0$ sector, so $\langle \sum_x G_x^2 \rangle = 0$ exactly; a gauge-symmetry-breaking error makes $\langle \sum_x G_x^2 \rangle > 0$ and the native falsifier fires, whereas a global charge check is comparatively blind. Gauss-law violation as a measurable diagnostic was established experimentally in an analog cold-atom simulator [10]; the digital lattice-Schwinger model is that treated by self-verifying simulation [9].

estimation validation [1]. Third, the positivity that underwrites the moment bounds is the positivity of a diagonal $T = 0$ Lehmann measure; off-diagonal or finite-temperature spectral functions are not positive measures in general and require separate treatment [14]. Fourth, our demonstrations use exact moments, an idealization; on hardware, moments carry statistical and systematic error, and the correct object is a robust interval-moment test whose falsification threshold accounts for that uncertainty [1]. We regard the exact-moment results as the theory limit that the hardware-robust version must recover; Appendix B gives a shot-budget forecast of that robust version calibrated on the exact state, with the fully device-side deployment—propagating measured rather than assumed bias—left as the remaining step.

Several extensions follow directly. An interval-moment formulation, propagating shot noise and mitigation bias into a two-sided falsification band, is forecast on the exact state in Appendix B; carrying it to a full device-side deployment—with the mitigation bias measured and bounded from error-model violation rather than assumed [118]—is the immediate priority [1]. The Gauss-law check demonstrates that the framework accepts any exactly known operator identity as a native falsifier, which invites a catalogue of domain-specific conditions—conservation laws, symmetry sectors, and sum rules—each screening a different error class [10, 89]. Extending positivity-based bounds to off-diagonal and finite-temperature spectra, where the naive Lehmann positivity fails, is the main theoretical opening [14]. Fi-

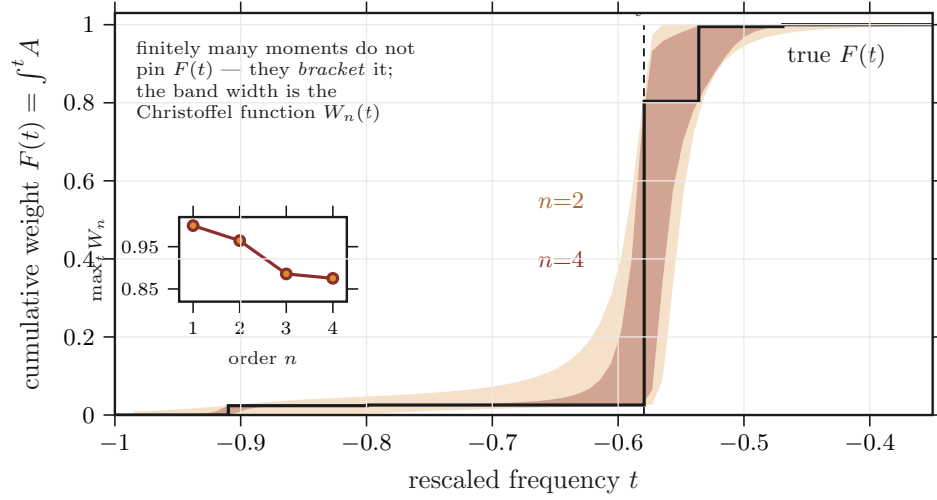


FIG. 8. **The bracketing mechanism behind the theorem.** Finitely many moments do not pin the spectrum—they *bracket* its cumulative weight $F(t)$ between the extremal Gauss–Radau (principal-representation) measures that share $m_0, \dots, m_{2n}$. The true $F(t)$ (ink staircase) lies inside every bracket; the vertical width of the band is the Christoffel function $W_n(t)$ (an equality at interior points of a continuous support; for the finite atomic measure plotted the drawn staircase width is grid-limited), and the $n=4$ band (deep) nests inside the $n=2$ band (light), so the certificate tightens with moment order (inset: $\max_t W_n$). At these low, transportable moments the bracket is still loose ($\max_t W_4 \approx 0.87$); a tight (~ 0.1) certificate needs ~ 20 – 30 moments. Doped Hubbard $L = 6$, $U/t = 8$ current-operator Lehmann measure, rescaled to $t \in [-1, 1]$ [11, 82, 83].

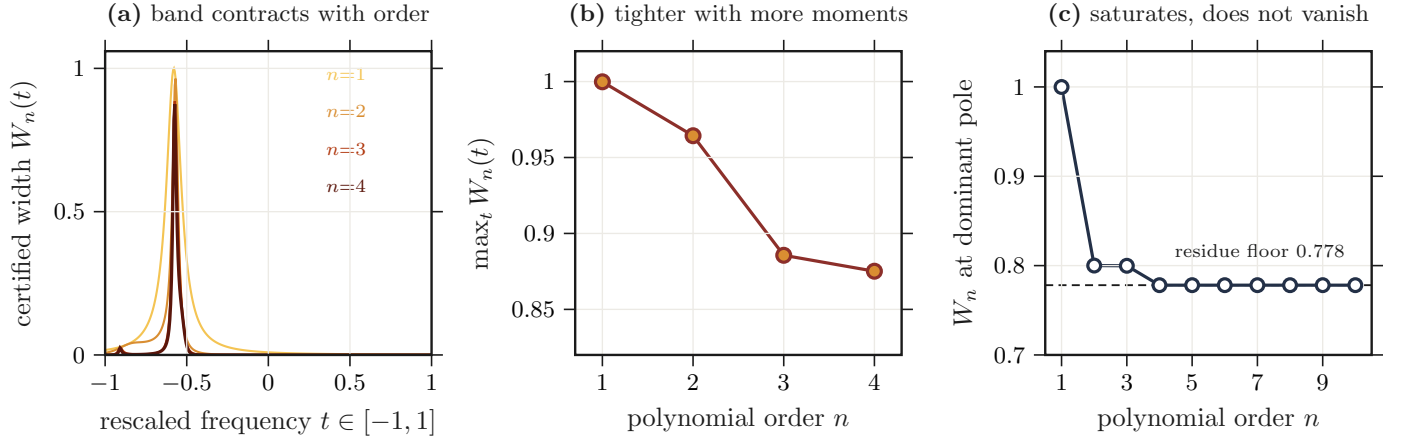


FIG. 9. **The certified cumulative-weight miss-distance.** (a) The Christoffel band width $W_n(t) = 1/\sum_{j=0}^n p_j(t)^2$ of the doped Hubbard current-operator Lehmann measure (rescaled to $t \in [-1, 1]$) contracts as the moment order n grows. (b) Its maximum over t shrinks monotonically, $W_{n+1} \leq W_n$ [11], from 0.9998 ($n=1$) to 0.875 ($n=4$) on a dense grid—a certified but still loose miss-distance at these low, transportable moments. (c) At the dominant spectral atom ($\omega = 11.09$, residue 0.778), W_n evaluated in high precision *saturates at the residue* and does *not* vanish: pointwise $W_n \rightarrow 0$ is a thermodynamic-limit statement for a densifying continuous density [13, 117], not a property of a finite atomic spectrum. The certified quantity is cumulative weight at interior points, not a line shape [11, 12].

nally, because the moment machinery is exactly that of the kernel-polynomial, recursion, and quadrature methods [11, 89, 90], the screen composes naturally with the sample-based diagonalization and Krylov pipelines that generate the spectra it tests [2, 5, 8, 44–46]. The through-line is unchanged: a quantum spectrum earns trust not by being checked against an answer we do not have, but by surviving the tests it was honestly given the chance

to fail [6].

A. Falsifying machine-learned reconstructions, and the quantum-for-AI outlook

Seen through the machine-learning lens introduced in the Introduction, the reconstruction step is a genera-

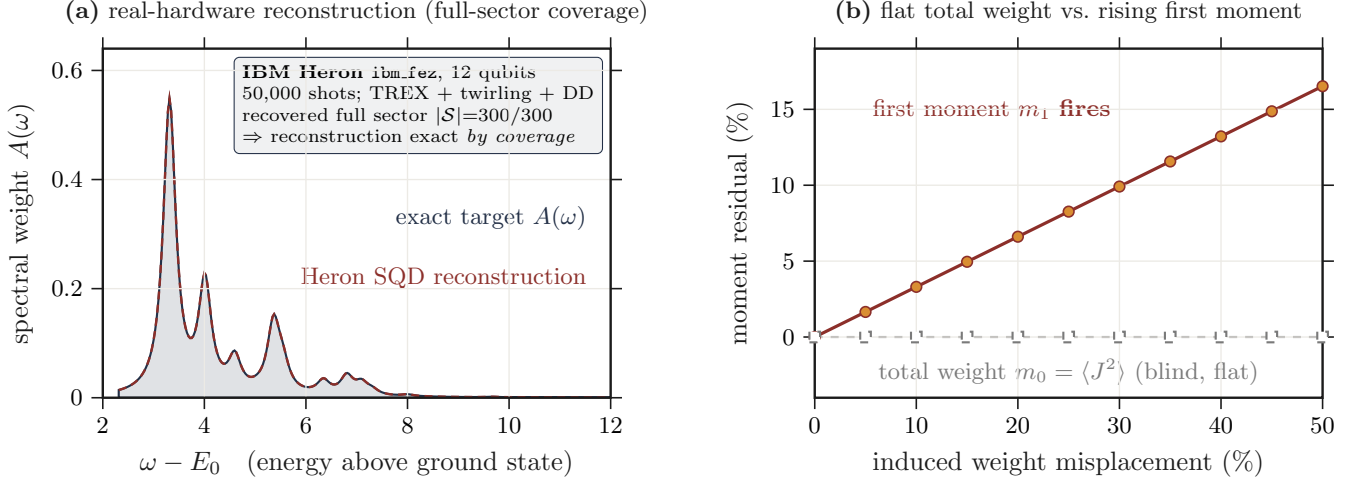


FIG. 10. Real IBM Heron demonstration. Applied to the sample-based-diagonalization spectral data of the companion study, measured on an IBM Heron processor. **(a)** At the demonstrated size— $L = 6$ Hubbard, $U/t = 4$, electron-addition sector—a real IBM Heron run (`ibm_fez`, 12 qubits, 50,000 shots, with TREX readout mitigation, Pauli twirling and dynamical decoupling) recovered the *full* $N \pm 1$ determinant sector ($|\mathcal{S}| = 300/300$), so the sample-based-diagonalization reconstruction coincides with the exact one *by coverage* (the markers are ω -nodes of that reconstruction, lying on the exact curve); the run establishes hardware executability and full-sector recovery, not a device-error line shape or a fidelity claim. **(b)** On that reconstruction the screen fires on a deliberately induced weight misplacement (first moment m_1) while the total-weight check (zeroth moment m_0) stays flat, reproducing the teeth-vs-blind contrast of Fig. 6. Baseline is exact-vs-exact; the sweep isolates firing on the induced error, not device noise—the exact-moment idealization still awaits its robust interval-moment form on noisy hardware [2, 5].

tive fit: a model of the spectral measure inferred from the device’s bitstring samples, of the kind studied under neural quantum states [15], neural-network quantum-state tomography [61], and flow- and policy-based generative samplers of physical distributions [62, 119]—the same role played by the generative decoders in our own reconstruction pipeline, whose landscape and classical-simulability frontier we survey in a companion review [63]. The complementary strategy, predicting many observables from few measurements without reconstructing the state, is the classical-shadows programme [72, 76], to which our estimator branch is the reconstruction-free counterpart; its spectral specialization, which extracts energy gaps from shadows without ever forming $A(\omega)$, is shadow spectroscopy [120], complementary to our test of an already-formed reconstruction. Against this backdrop the self-falsifying screen is a *necessary-condition falsification test for a machine-learned generative model of quantum-computed data*: the learned reconstruction is fit to the sampled distribution, and the screen then holds it to independent physical moment constraints—sum rules, conserved-charge identities, Hankel positivity—that it was never fit to, evaluated on the same samples and never read back off the model [3, 14]. This is an out-of-distribution physical-consistency check of a generative fit—a falsifiable guard against the characteristic generative-model failure mode of emitting a confident, plausible-but-wrong output (in the vocabulary of machine learning, a hallucination), of which the wrong reconstruction of Fig. 3, a valid-

looking Mott spectrum that misplaces weight, is a physical instance—decisive in one direction only: a model that violates a moment it did not train on is refuted, while one that respects it has survived a genuine opportunity to fail rather than been certified [6].

These considerations are not abstract: the pipelines the screen attaches to are shipping products, and it has concrete plug-in points in them. Sample-based diagonalization is a released component of quantum-centric supercomputing workflows [2, 45, 58], whose reconstructed spectra carry exactly the determinant-truncation weight-misassignment failure mode the screen targets [78, 80]; there the moment battery is a near-zero-cost post-processing validity gate—same-sample operator expectations, no additional circuits—attachable to any such module. On the hardware side, trapped-ion processors now emit $A(k, \omega)$ and $S(q, \omega)$ as advertised outputs [20, 43], and enterprise materials workflows emit $A(k, \omega)$ for users who cannot recompute it classically [59]; a reconstruction-independent sum-rule check is a self-consistency label such outputs can ship with. Because it needs only same-basis samples, the screen is moreover a vendor-agnostic validity primitive, running identically across superconducting, ion-trap, and neutral-atom backends, and it complements the application-oriented benchmarks that already serve as a cross-industry quality layer [56]—which score a device on a task class, whereas the screen instead tests whether one particular output is physically self-consistent—a necessary condition it can decisively fail but never prove. We are candid about the

limits of this relevance. The regime where such a check is indispensable—beyond-classical output with no cross-check, the endpoint of the fault-tolerant roadmap [60]—is not yet here, and a refutation-only instrument serves risk management and credibility rather than the advantage claims a vendor is structurally motivated to feature. The cautionary value is nonetheless concrete: a flagship hardware result that outside observers could not independently falsify on the evidence shown [121] is, by analogy and at the level of a device rather than a spectrum, exactly the situation a reconstruction-independent necessary-condition screen exists to forestall.

We place this within the broader programme of using quantum processors as inferential and generative engines only as an outlook, and with the boundary stated plainly. The prospect is a device whose sampled outputs are the data a learned model is built on, and the corresponding need is that those outputs be validated by falsifiable, reconstruction-independent tests rather than trusted because a model fits them; what this paper supplies is that validation layer. What it does *not* supply, and does not claim, is any demonstrated quantum learning advantage—of the kind now established on dedicated platforms [122]: at the accessible scales the sampled subspace is classically reproducible, and it is precisely because we do not rest on an advantage claim that the falsification contribution stands on its own [5]. The screen validates learned reconstructions of quantum-computed spectra; it neither asserts nor requires that the quantum computer learns better than a classical one.

A natural objection follows from that very framing. The sharpest quantum-learning separations are for functionals that are *nonlinear* in the state—purity, Rényi entropies, and related multi-copy quantities efficiently estimable only with a quantum memory [123, 124]—so if the reconstruction is a learned model, why screen it with linear moments $\hat{m}_k = \langle O_k \rangle$ rather than a nonlinear, two-copy correctness functional? The first answer is practical and settles the matter on present hardware: a two-copy estimator of a functional such as $\text{tr}(\rho^k O)$ carries its own replica variance and its own mitigation bias, both of which enter the falsification interval additively, widening the two-sided band on the `ibm_fez` budget of Appendix B well beyond the few-percent thresholds a single-copy moment already reaches; a refutation-only instrument gains nothing from an estimator whose interval it can only loosen. The second answer is structural, and we state it only to leading order and within a scoped regime. Model a smooth scalar functional $G(\rho)$ —a candidate correctness statistic, generally nonlinear in ρ —evaluated at the true state ρ^* (the correct spectral state), from which a hallucination displaces the fit by $\delta\rho$. Here ρ^* is the reference (null) state, not an extremum of G : were G stationary at ρ^* its first differential would vanish and only the second-order (Class II) case below would remain. A Taylor expansion gives a hallucination gap $\delta G = \langle DG[\rho^*], \delta\rho \rangle + O(\|\delta\rho\|^2)$ whose leading term is linear in the displacement. Two cases exhaust

the near-optimum picture. Either $DG[\rho^*] \neq 0$ and the gap is first-order, its entire leading signal being a single *linear* observable $O_{\text{eff}} = DG[\rho^*]$ —the quantum score, whose expectation is the locally-most-powerful test of “no hallucination” against a small displacement—so that nonlinear, multi-copy machinery adds nothing at this order; or $DG[\rho^*] = 0$, the gap is second-order and flat, and the displacement is undetectable by any first-order probe, linear or otherwise. This is the leading-order content of the paper’s own claim that independence is what gives the screen teeth: a probe fires iff the functional is non-stationary along the displacement, so independence $\Leftrightarrow$ non-stationarity $\Leftrightarrow$ detection power, and a linear observable already captures the whole first-order signal a hallucination can present. We are deliberate about what this argument is and is not. Its mathematics is folklore—the score statistic and the locally-most-powerful test are classical [125, 126], and the first-order reduction is ordinary linear-response reasoning [81]—so we claim novelty only for the *deployment* of this logic as a reconstruction-independent hallucination screen, not for the statistics. The identification of our O_k with $O_{\text{eff}} = DG[\rho^*]$ is an *analogy*, not an identity: ρ^* is unknown, and the O_k are fixed physical sum-rule generators rather than the gradient of any estimated objective; the claim is only that a *linear* observable is the right functional *form* for the leading signal, of which our moments are physically-motivated instances. The second, flat case is a *universal* undetectability limit—no probe of any kind, linear or nonlinear, single- or two-copy, sees a stationary displacement to first order—so it disadvantages no method and is a boundary we record, not a construction we offer. None of this is in tension with the genuine two-copy exponential advantages for nonlinear properties of adversarial or Haar-random states [76, 123, 124]: those live precisely in the regime our near-optimum, single-displacement picture excludes—far from any optimum, where a functional’s dependence on ρ is not captured by its first differential—and we make no claim on that territory. Finally, two guards against over-reading. This linear/first-order argument concerns nonlinearity *in the state* ρ and is unrelated to the Hankel–Stieltjes battery of Sec. V, which is nonlinear *in the moments* and which the screen already uses; the two must not be conflated. And we assert no provably optimal hallucination detector: the optimality is local, first-order, and conditional on the near-optimum, first-order-detectable regime (Class I), and it says nothing about second-order-flat hallucinations (Class II), which stay invisible to every method alike.

Two threads of this paper meet in a single epistemological point. A necessary condition has the logical form of a universal law—one violation refutes the spectrum outright, while conformity establishes nothing—and this asymmetry is forced on us, not chosen for elegance: confirming a spectrum reported beyond independent reach would require a trusted reconstruction the regime does not supply, and cannot be presumed to supply later,

since the frontier of classical reach itself shifts as classical methods improve [47, 53]. An instrument that can only refute is then the honest match to the epistemic situation rather than a concession. What gives such refutation evidential force is the *severity* of each test in Popper’s own sense—a test the spectrum would very probably have failed had it been wrong: a reconstruction-independent moment is severe precisely because its agreement was not built in, whereas a moment read back off $A(\omega)$ is a test of zero severity that cannot fail and so adds no weight however often it is passed [5, 6]. This is why survival of the battery is corroboration in the strict, deflationary sense and never a probability of correctness, which a necessary-but-not-sufficient screen cannot confer. The same asymmetry has an exact mathematical form: a learned reconstruction of a $T = 0$ Lehmann spectrum is a candidate solution to a truncated moment problem, so whether it may be trusted is decided not by a held-out fit but by membership in a Hankel-positive convex body, within the sharp and computable Christoffel radius of Sec. V, and thereafter corroborated—never proved [11, 82, 83]. The same noncommutative moment/positivity machinery has recently been turned into certified bounds on ground-state *observables* of many-body systems [127], a certify-a-property-not-the-answer lane adjacent to ours. As a quantum processor is used more as an inferential and generative engine, its reports come to resemble testimony one cannot independently recompute, warranted—if at all—not by re-derivation but by the physical constraints one can make them confront. The screen developed here is one instance of that discipline and only that: one component of the falsifiable-trust layer such a programme requires, confined to a single error class, certifying no correctness of the reconstruction, its conditions necessary and not sufficient, and asserting no quantum learning advantage [1, 14].

ACKNOWLEDGMENTS

Computations used exact diagonalization and matrix-product-state (TeNPy) tools; the hardware spectral data are those of the companion study [5]. Code and full reproduction materials—the exact sector-Lanczos engine, the interval-moment and joint-battery scripts, the per-figure data, and a master notebook that regenerates every figure and number end to end—are openly available at <https://github.com/nicolasbonilla/self-falsifying-quantum-spectroscopy>.

Appendix A: Derivation of the moment identities

We derive the operator form of the spectral moments used throughout, making explicit that each m_k is a ground-state expectation value computable without the dynamics. Let $H = H^\dagger$ have a non-degenerate ground state $|0\rangle$, $H|0\rangle = E_0|0\rangle$, and a complete orthonormal

eigenbasis $\{|n\rangle\}$ with $H|n\rangle = E_n|n\rangle$; let $J = J^\dagger$ be a Hermitian probe (a current or density). The diagonal zero-temperature Lehmann spectral function

$$A(\omega) = \sum_n |\langle n|J|0\rangle|^2 \delta(\omega - (E_n - E_0)) \quad (\text{A1})$$

is a non-negative measure supported on $\omega = E_n - E_0 \geq 0$, and its power moments are $m_k = \int \omega^k A(\omega) d\omega = \sum_n (E_n - E_0)^k |\langle n|J|0\rangle|^2$.

Writing $|\langle n|J|0\rangle|^2 = \langle 0|J|n\rangle\langle n|J|0\rangle$, using $\langle n|(H - E_0)^k = (E_n - E_0)^k \langle n|$, and closing with completeness $\sum_n |n\rangle\langle n| = \mathbb{1}$,

$$m_k = \sum_n \langle 0|J|n\rangle (E_n - E_0)^k \langle n|J|0\rangle = \langle 0|J (H - E_0)^k J|0\rangle. \quad (\text{A2})$$

The ground state is annihilated by $H - E_0$, so acting to the right $(H - E_0)J|0\rangle = [H, J]|0\rangle$, and iterating,

$$(H - E_0)^k J|0\rangle = \text{ad}_H^k(J)|0\rangle, \quad \text{ad}_H(J) \equiv [H, J], \quad (\text{A3})$$

where ad_H^k denotes the k -fold nested commutator $[H, [H, \dots [H, J]]]$. Substituting Eq. (A3) into Eq. (A2) gives the closed operator form

$$m_k = \langle 0|J \text{ad}_H^k(J)|0\rangle = \langle 0|J \underbrace{[H, [H, \dots [H, J]]]}_k |0\rangle \equiv \langle O_k \rangle, \quad (\text{A4})$$

with $O_k = J \text{ad}_H^k(J)$ built solely from H and J —no reference to the spectrum or to the reconstructed $A(\omega)$.

The lowest moments specialize immediately. For $k = 0$, Eq. (A4) gives the total weight $m_0 = \langle 0|J^2|0\rangle = \langle J^2 \rangle$. For $k = 1$, $m_1 = \langle 0|J[H, J]|0\rangle$; expanding the symmetric double commutator $[J, [H, J]] = 2JHJ - J^2H - HJ^2$ and using $H|0\rangle = E_0|0\rangle$,

$$\langle 0|[J, [H, J]]|0\rangle = 2\langle 0|J(H - E_0)J|0\rangle = 2m_1, \quad (\text{A5})$$

so that

$$m_1 = \frac{1}{2} \langle 0|[J, [H, J]]|0\rangle. \quad (\text{A6})$$

This is the first-moment (double-commutator) sum rule of A_{JJ} ; for the current operator J it fixes the mean frequency of the spectral weight. It is distinct from the optical-conductivity f -sum rule $\int_0^\infty \sigma_1 d\omega = \frac{\pi}{2} \langle -\hat{T} \rangle$, which is the m_{-1} moment of A_{JJ} (a ground-state kinetic energy) [3, 4, 7]. The symmetrized form (A6) is manifestly real and independent of the additive constant E_0 , which the one-sided expression $\langle 0|J[H, J]|0\rangle$ obscures.

Two features of Eq. (A4) make it the operational heart of the screen. First, O_k is an explicit operator polynomial in H and J ; expanded in the computational (occupation-number) basis, $\langle O_k \rangle$ becomes a weighted sum of basis observables whose ground-state expectation is estimated from the same sample set $\mathcal{S}$ that produced the reconstruction (Sec. II A)—this is the reconstruction-independent estimator $\hat{m}_k$. Second, the derivation used

only (i) the positivity of the measure (A1) and (ii) the ground-state annihilation $(H - E_0)|0\rangle = 0$; it therefore holds verbatim for any Hermitian probe and any Hamiltonian, which is the algebraic reason the screen transports across models.

For a non-Hermitian probe the same steps hold with $J^\dagger$ on the bra side, $m_k = \langle 0|J^\dagger(H - E_0)^k J|0\rangle$, so that $m_0 = \langle J^\dagger J \rangle$. The single-particle spectral function $A_{k\sigma}(\omega)$ of Fig. 3 is the two-sided (photoemission plus inverse-photoemission) spectrum of the fermionic probe $c_{k\sigma}$, and its moments close under *anticommutators* rather than the double commutator:

$$\begin{aligned} \int A_{k\sigma} d\omega &= \langle \{c_{k\sigma}, c_{k\sigma}^\dagger\} \rangle = 1, \\ \int \omega A_{k\sigma} d\omega &= \langle \{[c_{k\sigma}, H], c_{k\sigma}^\dagger\} \rangle = \epsilon_k + U\langle n_{-\sigma} \rangle, \end{aligned} \quad (\text{A7})$$

the Hartree-shifted band centroid [3]. The Hermitian double-commutator map above is the special case $J = J^\dagger$; Eq. (A7) is the version used for the single-particle screen of Fig. 3, and is likewise a ground-state expectation of explicit operators estimable from the same samples.

Appendix B: The reconstruction-independent moment estimator

We give the explicit estimator $\hat{m}_k$ and show precisely why it is independent of the reconstruction and why it—unlike the spectral moment $\bar{m}_k$ —detects subspace truncation. Sample-based diagonalization returns a set $\mathcal{S}$ of computational-basis determinants $\{|x\rangle\}$ and, from the classically diagonalized projected Hamiltonian $H_{\mathcal{S}} = \Pi_{\mathcal{S}} H \Pi_{\mathcal{S}}$ with $\Pi_{\mathcal{S}} = \sum_{x \in \mathcal{S}} |x\rangle\langle x|$, the ground state $|0\rangle = \sum_{x \in \mathcal{S}} c_x |x\rangle$ and eigenpairs $\{E_n, |n\rangle\}_{n \in \mathcal{S}}$ [2, 8]. The probe J and the Hamiltonian H are sparse in this basis, with known matrix elements $J_{yx} = \langle y|J|x\rangle$ and $H_{yx} = \langle y|H|x\rangle$.

The moment $m_k = \langle 0|J(H - E_0)^k J|0\rangle$ of Appendix A admits two evaluations that coincide only when $\mathcal{S}$ is complete for the operators involved. Inserting the *subspace* resolution $\Pi_{\mathcal{S}}$ in place of the identity yields the reconstruction (spectral) moment,

$$\bar{m}_k = \sum_{n \in \mathcal{S}} (E_n - E_0)^k |\langle n|J|0\rangle|^2 = \langle 0|J[\Pi_{\mathcal{S}}(H - E_0)\Pi_{\mathcal{S}}]^k J|0\rangle, \quad (\text{B1})$$

which is exactly $\int \omega^k A(\omega) d\omega$ and lives entirely inside $\mathcal{S}$. The reconstruction-independent estimator instead applies the sparse operator $O_k = J(H - E_0)^k J$ directly,

$$\hat{m}_k = \langle 0|J(H - E_0)^k J|0\rangle = \sum_{x, x' \in \mathcal{S}} c_x^* c_{x'} \langle x|J(H - E_0)^k J|x'\rangle, \quad (\text{B2})$$

where each matrix element $\langle x|O_k|x'\rangle$ is a finite product of sparse operators summing over intermediate determinants that need *not* lie in $\mathcal{S}$. This is the operational meaning of independence: $\hat{m}_k$ probes the action of J

and H on $|0\rangle$ beyond the sampled subspace, whereas $\bar{m}_k$ is confined to it.

The two agree exactly when the states generated from $|0\rangle$ by these operators— $J|0\rangle$, $(H - E_0)J|0\rangle$, and so on up to order k —already lie in $\text{span } \mathcal{S}$, so that $(\mathbb{1} - \Pi_{\mathcal{S}})$ annihilates them and $\Pi_{\mathcal{S}}$ may be dropped from Eq. (B1). Under determinant-subspace truncation this fails: writing $|\varphi\rangle = J|0\rangle$,

$$\hat{m}_k - \bar{m}_k = \langle \varphi|(H - E_0)^k |\varphi\rangle - \langle \varphi|[\Pi_{\mathcal{S}}(H - E_0)\Pi_{\mathcal{S}}]^k |\varphi\rangle, \quad (\text{B3})$$

the spectral weight the truncation leaks outside $\mathcal{S}$. The discrepancy $\Delta_k = |\hat{m}_k - \bar{m}_k|$ is thus nonzero precisely to the extent that $\mathcal{S}$ omits configurations carrying weight: the independent estimator fires, while the circular control compares $\bar{m}_k$ to itself and returns $\Delta_k \equiv 0$, staying blind (Sec. IV).

Equation (B2) is equivalently an importance-weighted average over $\mathcal{S}$,

$$\hat{m}_k = \sum_{x \in \mathcal{S}} |c_x|^2 O_k^{\text{loc}}(x), \quad O_k^{\text{loc}}(x) = \frac{\langle x|O_k|0\rangle}{\langle x|0\rangle}, \quad (\text{B4})$$

the local-operator estimator familiar from variational Monte Carlo [15]; the local moment $O_k^{\text{loc}}(x)$ again sums over determinants connected to x by O_k and so reaches outside $\mathcal{S}$. Because $\mathcal{S}$ and the amplitudes c_x are fixed by a finite number of measured shots and by the error-mitigation pipeline, $\hat{m}_k$ carries statistical and systematic uncertainty δ_k , and the exact-moment falsifier $\Delta_k > 0$ is replaced on hardware by the robust interval-moment test $\Delta_k > \delta_k$ anticipated in the main text [1].

This is the concrete form of the generative-model falsifier of the Discussion: $\hat{m}_k$ evaluates a physical constraint using operator support that the learned reconstruction—fit to the sampled distribution on $\mathcal{S}$ —may have omitted, so a model that has misplaced spectral weight cannot satisfy it unless the omitted physics was in fact absent. It is an out-of-distribution test in the precise sense that it interrogates the model exactly where it did not fit.

The exact-moment falsifier acquires its hardware form here. Modelling the estimator uncertainty as $\delta_k^2 = \text{Var}(O_k)/N_s + b_k^2$ (shot variance over N_s samples plus a mitigation-bias budget b_k), the two-sided decision is to falsify when $|\hat{m}_k - \bar{m}_k| > z_{1-\alpha/2} \delta_k$ at level α . The *battery-level version* is a semidefinite feasibility test: the spectrum survives if there exists a moment vector $\mu_\ell \in [\hat{m}_\ell - \delta_\ell, \hat{m}_\ell + \delta_\ell]$ whose Hankel matrix is positive semidefinite, $H_n \succeq 0$ (and, for the one-sided $T = 0$ support, the shifted $H_n^{(1)} \succeq 0$); infeasibility falsifies. This is the robust interval-moment test anticipated in the main text [1]; the per-demonstration operators, targets, and intervals accompany the reproduction materials.

We close with a shot-budget-informed forecast of this test at the real `ibm_fez` scale and budget ($N_s = 5 \times 10^4$), computed as a model on the exact state—not a device measurement—because the device bitstring counts of the companion run were not retained. Two inputs are therefore modelled, both in the safe direction. The shot term

uses the exact-state local-estimator variance $\text{Var}(O_k) = \langle O_k^2 \rangle - \langle O_k \rangle^2$, a *lower bound* on the true per-shot uncertainty (which additionally carries ratio-estimator and Pauli-grouping overhead); the bias term b_k is a residual we *assume and sweep* rather than measure. For the doped $L = 6$, $U/t = 4$ current probe, $\text{Var}(O_1) \approx 3.2 \times 10^2$ gives a shot standard error $\sqrt{\text{Var}(O_1)/N_s} \approx 0.081$ (1.9% of m_1); combined with a 2% bias budget the 95% two-sided falsification threshold is $z_{0.975} \delta_1 \approx 0.23$ (5.4% of m_1). A principled, budget-informed threshold thus lands near the 5% figure a practitioner would pick by hand, corroborating that choice rather than replacing it. Within this threshold the forecast corroborates the converged reconstruction ($|\hat{m}_1 - \bar{m}_1| \sim 10^{-15}$) and refutes reconstructions truncated below $\sim 90\%$ determinant coverage. Read alongside the teeth figure, which first fires near $\sim 7\%$ retained determinants ($d \approx 1.6 \times 10^4$ of 2.4×10^5 ; Fig. 6), this $\sim 90\%$ figure reflects not a sharper test but a different one: Fig. 6 is the large $L = 12$ sector scored by a *relative* discrepancy $|\hat{m}_1 - \bar{m}_1|/\hat{m}_1$ against a fixed fractional line, whereas the present forecast is the small $L = 6$ sector scored by an *absolute* shot-budget interval $z \delta_1$ tied to the `ibm_fez` budget—so the two coverage percentages are set against incommensurate rulers on different-sized sectors, not one test yielding two answers. The direction of the residual uncertainty is favourable: a wider true threshold makes a refutation-only instrument strictly *more* conservative—fewer, safer rejections—so underestimating δ_k errs safe, and we state the sensitivity plainly, since at twice the shot floor or a 4% bias the lone first moment passes some still-truncated reconstructions.

That fragility is a property of a single moment. It has an accidental blind spot—one intermediate truncation whose misplaced weight leaves m_1 coincidentally intact within its interval—and the joint battery (m_0, m_1, m_2) with the Hankel–Stieltjes feasibility test closes *that* blind spot: the second moment, weighing the spectral tail the first averages over, refutes the same reconstruction by many times its interval ($|\hat{m}_2 - \bar{m}_2| \approx 3.3$ against a threshold ≈ 1.8) and keeps refuting at the doubled bias budget where m_1 alone goes blind. We report this as closing the blind spot *exhibited here*, not as a proof that the joint battery is free of blind spots in general—no finite battery can be, since necessary conditions are not sufficient. Two boundaries frame the forecast. The discrepancy is measured against the exact oracle $\hat{m}_k$, so it *calibrates* the interval where classical ground truth exists; the fully device-side version, propagating measured rather than assumed bias, is the deployment step. And where the finite-shot moment-constraint machinery of Refs. [127, 128] uses feasibility to *certify* that a static property holds, we use *infeasibility* to *refute* a reconstruction against constraints it was never fit to—the screen falsifies and never certifies—applied to a frequency-resolved spectral (Lehmann) measure rather than an expectation value or steady-state property. The classical spectral-moment sum rules themselves are textbook [3, 89]; what is new is only their use as reconstruction-independent,

interval-valued refutation constraints on a hardware-computed spectral function.

Appendix C: Proof of the Christoffel miss-distance theorem

We prove the equality of the extremal cumulative-weight spread with the Christoffel function through three classical lemmas [11, 83, 84]. Write $K_n(x, y) = \sum_{j=0}^n p_j(x)p_j(y)$ for the Christoffel–Darboux kernel and $\lambda_n(t) = 1/K_n(t, t)$ for the Christoffel function, and assume the moment vector is interior to the moment cone, $H_n = [m_{i+j}]_{i,j=0}^n \succ 0$ (for a finite N -atom Lehmann spectrum this holds for $n \leq N - 1$).

Lemma 1 (variational identity). $\lambda_n(t) = \min\{\int |P|^2 d\mu : P \in \mathcal{P}_n, P(t) = 1\}$, with minimizer $P^* = K_n(\cdot, t)/K_n(t, t)$. Expanding $P = \sum_{j=0}^n c_j p_j$ gives $\int |P|^2 d\mu = \sum_j c_j^2$ while $1 = P(t)^2 = (\sum_j c_j p_j(t))^2 \leq (\sum_j c_j^2) K_n(t, t)$ by Cauchy–Schwarz, with equality at $c_j = p_j(t)/K_n(t, t)$; the minimum is $1/K_n(t, t) = \lambda_n(t)$.

Lemma 2 (maximal atom). Among positive measures ν with $\int \omega^k d\nu = m_k$ for $k \leq 2n$, the largest mass at a point t is $\max_\nu \nu(\{t\}) = \lambda_n(t)$. For the extremal P^* of Lemma 1, $\nu(\{t\}) = P^*(t)^2 \nu(\{t\}) \leq \int (P^*)^2 d\nu = \int (P^*)^2 d\mu = \lambda_n(t)$, the middle equality using $\deg(P^*)^2 = 2n$ and that ν shares those moments; the bound is attained by the $(n+1)$ -point Gauss–Radau rule with a node prescribed at t , a positive measure matching $m_0, \dots, m_{2n}$ whose weight at t is $\lambda_n(t)$.

Lemma 3 (ordinate gap). For t interior to $\text{supp } \mu$, one-sided polynomial majorant/minorant $Q_\pm \in \mathcal{P}_{2n}$ with $Q_- \leq \mathbf{1}_{(-\infty, t]} \leq Q_+$ and $\int (Q_+ - Q_-) d\mu = \lambda_n(t)$ bracket $F_\nu(t) = \int \mathbf{1}_{(-\infty, t]} d\nu$ between $\int Q_- d\mu$ and $\int Q_+ d\mu$ for every moment-matching ν ; the bracket width is $\lambda_n(t)$, attained by shifting the maximal atom of Lemma 2 across t [83, 84]. Hence $\sup_{\nu_1, \nu_2} |F_{\nu_1}(t) - F_{\nu_2}(t)| = \lambda_n(t)$.

Combining the three, the extremal spread equals $\lambda_n(t) = W_n(t)$, and for any single reconstruction consistent with the moments $|F_{\text{rec}}(t) - F_{\text{true}}(t)| \leq W_n(t)$ everywhere; at edges or across a gap of $\text{supp } \mu$ the equality weakens to this upper bound. Monotonicity $W_{n+1} \leq W_n$ follows from $K_{n+1} = K_n + p_{n+1}^2 \geq K_n$.

Appendix D: Evaluating the certificate: from moments to W_n

The Christoffel function is computed from the moments $\{m_k\}_{k=0}^{2n}$ by a short, numerically stable pipeline [11, 86]. With the inner product $\langle f, g \rangle_\mu = \int f(\omega)g(\omega) d\mu(\omega)$, so that $\langle \omega^a, \omega^b \rangle_\mu = m_{a+b}$, the orthonormal polynomials follow from Gram–Schmidt on $\{1, \omega, \dots, \omega^n\}$ with $p_0 = 1/\sqrt{m_0}$ and obey the three-term recurrence

$$\beta_{j+1} p_{j+1}(\omega) = (\omega - \alpha_j) p_j(\omega) - \beta_j p_{j-1}(\omega), \quad (\text{D1})$$

with $\alpha_j = \langle \omega p_j, p_j \rangle_\mu$ and $\beta_j = \langle \omega p_j, p_{j-1} \rangle_\mu$, all fixed by $\{m_k\}$; this Stieltjes recurrence is numerically preferable to inverting the Hankel matrix directly. The nodes and weights of the $(n+1)$ -point Gauss rule are the eigenpairs of the Jacobi matrix $\mathbf{J}_{n+1} = \text{tridiag}(\beta_j, \alpha_j, \beta_j)$ (Golub–

Welsch), and the kernel on the diagonal, $K_n(t, t) = \sum_{j=0}^n p_j(t)^2$ evaluated by the same recurrence, yields the certificate $W_n(t) = 1/K_n(t, t)$. Every quantity in the guarantee is thus an explicit, auditable function of the measured moments.

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